Sahayak

Sahayak is a low-bandwidth welfare scheme discovery assistant built for NSS Open Projects 2026 - Track 1.1: AI-Based Multilingual Chatbot for Welfare Scheme Awareness.

The prototype helps a user fill a short eligibility profile, discover relevant welfare schemes, ask grounded follow-up questions, and generate a document checklist in English or Hindi.

Live demo: https://parzival1821.github.io/Multilingual_Chatbot/

Project Snapshot

- Status: Functional MVP deployed on GitHub Pages.
- Scope: 8 high-impact central welfare schemes.
- Languages: English and Hindi, with Hindi rendered in Devanagari.
- Core AI pattern: Retrieval-grounded answer generation over a curated scheme knowledge base.
- User flow: 4-6 simple eligibility signals produce a personalized scheme shortlist.
- Safety: Unknown questions return a fallback instead of hallucinated scheme details.
- Verification: Local core and static smoke tests pass with npm run check.

NSS Challenge Fit

Official expectation	Current implementation
Focused catalogue of 8-10 schemes	8 schemes included
Eligibility-questioning flow	Guided profile form with area, occupation, gender, income, land, housing, LPG, bank, a
Personalized shortlist	Ranked recommendations with match reasons
Multilingual interface	English + Hindi Devanagari
Document checklist	Copyable and downloadable scheme-specific checklist
Grounded answers	Retrieval over local scheme data with source links
Low-bandwidth thinking	Static app, text-first UI, no heavy framework, downloadable .txt handoff
Honest unknown handling	Safe fallback for unsupported questions

Scheme Catalogue

Scheme	Primary beneficiary	Current verifi
PM-KISAN Samman Nidhi	Landholding farmer families	Verified offici
Ayushman Bharat PM-JAY	Poor and vulnerable families needing health cover	Verified offici
Pradhan Mantri Ujjwala Yojana	Adult women from poor households without LPG	Verified offici
Pradhan Mantri Awas Yojana - Urban 2.0	Urban EWS/LIG/MIG families without a permanent house	Verified offici
National Social Assistance Programme	BPL elderly, widows, disabled persons, and vulnerable households	Verified offici
PM Street Vendor's AtmaNirbhar Nidhi	Urban and peri-urban street vendors	Official sourc
Atal Pension Yojana	Informal/unorganised workers with bank accounts	Official sourc
Sukanya Samriddhi Account	Girl child, through parent or guardian	Official sourc

System Architecture

Diagram: Rendered in the Visual Diagrams PDF.

User Journey

Diagram: Rendered in the Visual Diagrams PDF.

Grounded Answer Flow

Diagram: Rendered in the Visual Diagrams PDF.

Demo Personas

Diagram: Rendered in the Visual Diagrams PDF.

Implementation Details

- index.html and styles.css implement the responsive single-page dashboard.
- src/data.js stores curated scheme data, Hindi names, summaries, labels, documents, benefits, and source links.
- src/core.js contains eligibility scoring, retrieval, profile inference, checklist formatting, and grounded answer composition.
- src/app.js connects the UI to the recommendation engine, assistant, language switcher, and checklist actions.
- tests/core.test.js validates recommendation ranking, retrieval, fallback behavior, Hindi queries, and checklist logic.
- tests/static-smoke.test.js validates the static shell, deployed asset references, Hindi Devanagari copy, and Pages workflow shape.

Submission Documents

The submission/ folder contains Google-Docs-ready Markdown files for the official NSS deliverables:

- Combined PDF Submission Packet
- Problem Analysis Report
- Solution Framework
- Prototype and Implementation Plan
- Impact Projection
- Pilot Test Report
- Final Presentation Content
- Submission Portal Details
- Visual Diagrams

Run Locally

npm run dev

Then open:

http://localhost:4173

Test

npm test

Run all local checks:

npm run check

Demo Script

1. Open the live app or run it locally.
2. Click the Farmer preset.
3. Show PM-KISAN as a strong match and PM-JAY as an additional support option.
4. Click View documents on PM-KISAN and show the checklist panel.
5. Copy or download the checklist to demonstrate low-bandwidth handoff.
6. Ask: What documents do I need for Ayushman Bharat?

7. Switch to Hindi and ask: मुझे घर के लिए कौन सी योजना मिल सकती है?
8. Show source links, verification labels, and the safe fallback for unsupported questions.

Verification Checklist

- Farmer persona ranks PM-KISAN first.
- Woman-led household persona shows Ujjwala, PMAY-U, and Sukanya.
- Street vendor persona ranks PM SVANidhi first.
- Hindi mode uses Devanagari labels, prompts, scheme names, and checklist content.
- Recommendation View documents action scrolls to and highlights the checklist panel.
- Checklist copy and download actions produce a scheme-specific document list.
- Unsupported questions produce a safe fallback instead of invented scheme details.
- npm run check passes.

Project Positioning

Sahayak is intentionally built as a practical AI product rather than a custom ML research model. Its strength is the combination of structured eligibility logic, retrieval-grounded answers, bilingual UX, source-backed recommendations, and a low-bandwidth document handoff compatible with WhatsApp, SMS, or IVR channels.

Submission Documents

This folder contains Google-Docs-ready submission material for Sahayak, prepared for NSS Open Projects 2026 - Track 1.1: AI-Based Multilingual Chatbot for Welfare Scheme Awareness.

These files are written in Markdown so they render cleanly on GitHub and can be copied into Google Docs for final formatting.

Document Index

File	Purpose
01-problem-analysis-report.md	Problem framing, root causes, stakeholders, personas, and success criteria
02-solution-framework.md	Architecture, process flow, tech stack, safety, scalability, and risk mitigation
03-prototype-implementation-plan.md	Current MVP, feature map, deployment, testing, and validation evidence
04-impact-projection.md	Quantified impact model, assumptions, scenarios, and sustainability pathway
05-pilot-test-report.md	Pilot test report with scripted personas, test results, and validation metrics
06-final-presentation.md	8-slide final presentation content with speaker notes
07-submission-portal-details.md	Exact portal fields, links, and final submission checklist
08-visual-diagrams.md	Architecture, user journey, persona mapping, grounded answer, and handoff diagrams

Submission Links

- Live demo: https://parzival1821.github.io/Multilingual_Chatbot/
- GitHub repository: https://github.com/parzival1821/Multilingual_Chatbot
- Problem statement: Track 1 - AI & Intelligent Solutions, Challenge 1.1 - AI-Based Multilingual Chatbot for Welfare Scheme Awareness

PDF Exports

- Combined PDF packet: pdf/sahayak-submission-pack.pdf
- Individual PDF exports are available in pdf/.

Official Deliverable Coverage

Official deliverable	Repo artifact
Problem Analysis Report	01-problem-analysis-report.md
Solution Framework	02-solution-framework.md
Prototype / Implementation Plan	03-prototype-implementation-plan.md + live app
Final Presentation	06-final-presentation.md
Functional chatbot prototype	Live demo + source code
User-flow diagram covering 3 personas	README Mermaid diagrams + 08-visual-diagrams.md
Pilot test report	05-pilot-test-report.md
2-page impact projection	04-impact-projection.md

Deadline

Official submission deadline: 13 June 2026, 6:00 PM IST.

Problem Analysis Report

Project: Sahayak - Welfare Scheme Discovery Assistant
Track: Track 1 - AI & Intelligent Solutions
Challenge: 1.1 - AI-Based Multilingual Chatbot for Welfare Scheme Awareness
Live demo: https://parzival1821.github.io/Multilingual_Chatbot/

Executive Summary

India has a large welfare ecosystem, but eligible beneficiaries often fail to discover, understand, or complete applications for the schemes they qualify for. The official NSS challenge brief identifies awareness gaps, language barriers, low literacy, intimidating documentation, and low-bandwidth access as key bottlenecks. Sahayak addresses this by offering a low-bandwidth, multilingual assistant that asks simple eligibility questions, returns a personalized scheme shortlist, cites official sources, and generates a document checklist that can be copied or downloaded.

The MVP focuses on depth and trust rather than breadth. It covers 8 high-impact schemes and demonstrates the core product loop: profile collection, eligibility matching, grounded follow-up answers, and document handoff.

Problem Statement

Rural and low-income users face high friction when trying to identify and apply for welfare schemes. Even when schemes exist and funds are allocated, the last-mile user may not know:

- Which schemes apply to their household.
- What eligibility conditions matter.
- Which documents are required.
- Where to apply or verify details.
- Whether the information they receive is reliable.

The challenge is not only a search problem. It is a comprehension, trust, language, and process problem.

Context From Official Brief

The NSS Challenge 1.1 brief highlights:

- India runs more than 950 central and state welfare schemes.
- Awareness among eligible rural beneficiaries is low.
- Government portals often assume smartphone literacy and stable connectivity.
- Language barriers and documentation complexity reduce successful applications.
- A useful solution should be low-bandwidth, multilingual, grounded in official scheme data, and deployable to channels such as WhatsApp, SMS, or IVR.

Root Causes

1. Information Fragmentation

Scheme details are spread across multiple portals, PDFs, FAQs, state offices, and local intermediaries. Users must know the scheme name before they can search for details.

2. Eligibility Confusion

Eligibility is usually described in policy language. A beneficiary thinks in terms of life context: ”I am a farmer”, ”I do not have a gas connection”, ”I am a widow”, or ”I need help for my daughter's future.” Existing portals often do not translate that context into scheme matches.

3. Language and Literacy Barriers

Even when Hindi is available, content may be dense, bureaucratic, or partly in English. Users need simple language, regional-language support, and short explanations.

4. Documentation Friction

Users may know a scheme exists but fail to apply because they do not know what documents to prepare. The checklist is often the point where awareness becomes action.

5. Trust and Hallucination Risk

AI systems can produce confident but incorrect answers. For welfare schemes, wrong eligibility guidance can waste time, money, and trust. A responsible assistant must cite official sources and admit when it does not know.

6. Low-Bandwidth Constraints

Many target users have shared phones, limited data, unstable connectivity, or low-end devices. A text-first flow with copyable/downloadable output is more practical than a heavy app.

Stakeholders

Stakeholder	Need
Rural beneficiary	Simple scheme discovery and document checklist
Woman-led household	Awareness of LPG, housing, social security, and girl-child support
Street vendor / informal worker	Credit, pension, health, and social protection guidance
NGO field worker	Fast beneficiary triage during outreach camps
NSS volunteer	A repeatable tool for community scheme-awareness drives
District administration	Higher uptake without creating new welfare programs

Target Personas

Persona 1: Farmer

- Rural household.
- Owns cultivable land.
- Has a bank account.
- Needs income support and health protection.
- Relevant schemes: PM-KISAN, PM-JAY.

Persona 2: Woman-led Household

- Urban low-income household.
- Adult woman, no LPG connection.
- May be widowed or guardian of a girl child.
- Relevant schemes: Ujjwala, PMAY-U, Sukanya Samriddhi, NSAP depending on profile.

Persona 3: Street Vendor

- Urban or peri-urban informal worker.
- Has a bank account.
- Needs livelihood credit and future pension support.
- Relevant schemes: PM SVANidhi, Atal Pension Yojana, PM-JAY.

User Needs

Users need a system that can:

1. Ask only a few simple questions.
2. Convert life context into scheme recommendations.
3. Explain why a scheme is relevant.

4. Provide document checklists in the user's language.
5. Cite official sources.
6. Avoid unsupported claims.

Existing Alternatives and Gaps

Existing path	Gap
Government scheme portals	Accurate but difficult to navigate for first-time users
Search engines	Require knowing scheme names and reading long pages
Local intermediaries	Can help, but quality and availability vary
Generic chatbots	Risk hallucination if not grounded in official scheme data
PDF awareness drives	Static, not personalized, and hard to update

Design Requirements Derived From Analysis

- Focus on 8-10 high-impact schemes for quality over breadth.
- Keep eligibility questions simple.
- Support at least English and Hindi.
- Provide source-backed answers.
- Generate copyable/downloadable document checklists.
- Use a low-bandwidth static web implementation for MVP.
- Add a fallback for unsupported questions.
- Structure the code so additional schemes and languages can be added without redesigning the app.

Success Criteria

Criteria	MVP target
Scheme coverage	8 high-impact schemes
Language support	English + Hindi Devanagari
Completion flow	User can go from profile to shortlist to checklist
Answer safety	Unsupported questions return fallback
Source trust	Scheme cards and answers include official source links
Performance	Static site loads quickly on low-end connections
Repeatability	Tests validate recommendation and retrieval behavior

Conclusion

Sahayak reframes welfare awareness as a last-mile product problem. The core insight is that beneficiaries do not need a database dump; they need a short, trusted, language-aware decision flow. A focused, grounded, low-bandwidth assistant can help users move from uncertainty to action by identifying relevant schemes and preparing the documents needed to apply.

Solution Framework

Project: Sahayak - Welfare Scheme Discovery Assistant
Challenge: AI-Based Multilingual Chatbot for Welfare Scheme Awareness

Solution Overview

Sahayak is a static, low-bandwidth web assistant that combines structured eligibility scoring with retrieval-grounded follow-up answers. The system does not attempt to cover every welfare scheme. Instead, it focuses on a curated catalogue of 8 high-impact schemes and demonstrates a trustworthy product loop:

1. Collect simple profile signals.
2. Rank relevant schemes.
3. Explain why each scheme matches.
4. Answer follow-up questions using scheme-grounded retrieval.
5. Generate a document checklist for application preparation.

Product Flow

Diagram: Rendered in the Visual Diagrams PDF.

Technical Architecture

Diagram: Rendered in the Visual Diagrams PDF.

Tech Stack

Layer	Implementation
Frontend	HTML, CSS, vanilla JavaScript
Data	Curated local JSON-like objects in src/data.js
Recommendation engine	Rules-based scoring in src/core.js
Retrieval	Keyword/token matching over scheme names, tags, summaries, Hindi names, documents, and sea
Language support	English + Hindi Devanagari copy maps
Deployment	GitHub Pages
Testing	Node-based core and static smoke tests

Knowledge Base

The MVP includes 8 schemes:

- PM-KISAN Samman Nidhi.
- Ayushman Bharat PM-JAY.
- Pradhan Mantri Ujjwala Yojana.
- Pradhan Mantri Awas Yojana - Urban 2.0.
- National Social Assistance Programme.
- PM Street Vendor's AtmaNirbhar Nidhi.
- Atal Pension Yojana.
- Sukanya Samriddhi Account.

Each scheme entry stores:

- English name.
- Hindi display name.
- English and Hindi summaries.
- Target beneficiary.
- Benefits.

- Eligibility signals.
- Documents.
- Application paths.
- Official source URLs.
- Tags and search text.
- Verification status.

Recommendation Logic

The recommendation engine converts a user profile into a ranked shortlist. It evaluates signals such as:

- Rural vs urban area.
- Occupation.
- Gender.
- Age group.
- Income category.
- Cultivable land.
- Permanent house ownership.
- LPG connection.
- Bank account.
- Special categories such as widow, disability, and girl-child guardian.

Each scheme has simple scoring rules. For example:

- PM-KISAN gains score for farmer occupation, cultivable land, and rural area.
- Ujjwala gains score for adult woman, no LPG connection, and low income.
- PMAY-U gains score for urban household, no permanent house, and EWS/LIG/MIG/BPL income category.
- NSAP gains score for BPL status, senior citizen, widowhood, or disability.

Retrieval-Grounded Answering

The assistant uses retrieval instead of free-form hallucination:

1. Normalize the user's question.
2. Tokenize English and Devanagari text while preserving Hindi matras.
3. Match against scheme name, Hindi name, summary, documents, tags, and search text.
4. Check whether the match has enough context.
5. Compose an answer using only stored scheme data.
6. Attach an official source link.
7. Return a safe fallback if the match is weak.

Safety and Trust

Sahayak avoids unsupported claims through:

- Official source links on recommendation cards.
- Source links inside assistant answers.
- Verification labels for each scheme entry.
- Explicit recheck labels for schemes where latest rules should be verified.
- Safe fallback for unsupported questions.
- Tests for fallback behavior and high-risk query routing.

Multilingual Design

The MVP supports:

- English interface.
- Hindi interface in Devanagari.
- Hindi scheme display names.

- Hindi summaries, reasons, document names, benefits, and prompts.
- Devanagari query handling for examples such as:
  - आयुष्मान के लिए कौन से दस्तावेज चाहिए?
  - क्या मैं पीएम-किसान के लिए पात्र हूं?
  - मैं 55 वर्ष की विधवा हूं, मेरे लिए कौन सी योजना बेहतर है?

Low-Bandwidth Design Choices

- Static website with no frontend framework.
- Text-first UI.
- No heavy media dependencies.
- Copyable and downloadable .txt checklists.
- Works as a single-page app hosted on GitHub Pages.
- Compatible with WhatsApp/SMS/IVR channels because the logic is text-based.

Scale Readiness

Area	Readiness
Web deployment	Static GitHub Pages prototype with 8 schemes
Language expansion	Copy maps and data fields support additional languages
Scheme expansion	Data model separates scheme content from UI logic
Messaging channels	Text-first engine can be connected to WhatsApp/SMS gateways
Field operations	Checklist output supports volunteers during outreach camps

Risks and Mitigations

Risk	Mitigation
Outdated scheme rules	Verification labels and official source links
Hallucinated answers	Retrieval-grounded composition and fallback
Low literacy	Short questions, simple summaries, Hindi labels
Poor connectivity	Static low-bandwidth web app and text checklist
Excessive scheme scope	Focused 8-scheme catalogue
Language gaps	Copy maps and searchable Hindi display names

Financial and Operational Model

For MVP deployment:

- Hosting cost: free through GitHub Pages.
- Data storage: local static files.
- Maintenance: periodic scheme updates from official portals.
- Field deployment: NSS volunteers or NGO field workers can use it during awareness drives.

For WhatsApp/SMS deployment:

- Twilio/WhatsApp Business API costs depend on message volume.
- A district or NGO deployment can start with a small curated scheme catalogue and expand.
- The assistant can be maintained by a small technical volunteer team.

Conclusion

Sahayak is feasible because it avoids unnecessary complexity. It demonstrates the core AI product value: grounded personalization, multilingual access, and actionable document preparation. Its architecture is simple enough for a student team to build and maintain, while still extensible toward real last-mile channels.

Prototype and Implementation Plan

Project: Sahayak - Welfare Scheme Discovery Assistant
Live demo: https://parzival1821.github.io/Multilingual_Chatbot/
Repository: https://github.com/parzival1821/Multilingual_Chatbot

Prototype Status

The current prototype is a deployed static web app. It demonstrates the end-to-end user journey required for the NSS Challenge 1.1 MVP:

1. Select English or Hindi.
2. Fill an eligibility profile.
3. View personalized welfare scheme recommendations.
4. Ask grounded follow-up questions.
5. Generate, copy, and download a document checklist.

MVP Feature Map

Feature	Status	Notes
Live prototype	Complete	GitHub Pages deployment
8-scheme catalogue	Complete	Central schemes with official source links
Eligibility questionnaire	Complete	Uses household and occupation signals
Personalized recommendations	Complete	Ranked cards with reasons
Hindi interface	Complete	Devanagari UI, prompts, scheme names, summaries
Follow-up assistant	Complete	Retrieval-grounded answer generation
Safe fallback	Complete	Prevents unsupported answers
Document checklist	Complete	Copy/download text checklist
Demo personas	Complete	Farmer, woman-led household, street vendor
Automated tests	Complete	Core and static smoke checks

Repository Structure

```
.
├── index.html
├── styles.css
├── package.json
├── README.md
├── src
│   ├── app.js
│   ├── core.js
│   └── data.js
├── tests
│   ├── core.test.js
│   └── static-smoke.test.js
└── submission
    ├── 01-problem-analysis-report.md
    ├── 02-solution-framework.md
    ├── 03-prototype-implementation-plan.md
    ├── 04-impact-projection.md
    ├── 05-pilot-test-report.md
    ├── 06-final-presentation.md
    ├── 07-submission-portal-details.md
    └── 08-visual-diagrams.md
```

Key Files

File	Responsibility
index.html	Dashboard structure
styles.css	Responsive visual design
src/data.js	Scheme knowledge base and language labels
src/core.js	Recommendation, retrieval, profile inference, checklist formatting
src/app.js	UI event handling and rendering
tests/core.test.js	Behavioral tests for recommendations, assistant, checklist
tests/static-smoke.test.js	Static app and deployment checks

Implementation Flow

Diagram: Rendered in the Visual Diagrams PDF.

Local Run Instructions

npm run dev

Then open:

http://localhost:4173

Test Instructions

Run core tests:

npm test

Run all checks:

npm run check

Expected result:

All core tests passed.
Static smoke checks passed.

Deployment

The project is deployed using GitHub Pages from the main branch.

Deployment workflow:

1. Push changes to main.
2. GitHub Actions builds the static site.
3. Pages serves the app at the live demo URL.

Demo Script

1. Open the live app.
2. Use English mode first.
3. Click the Farmer preset.
4. Show PM-KISAN as a strong match.
5. Click View documents.
6. Copy/download the checklist.
7. Ask: What documents do I need for Ayushman Bharat?
8. Switch to Hindi.
9. Ask: मुझे घर के लिए कौन सी योजना मिल सकती है?
10. Show Hindi recommendations, source links, and fallback behavior.

Prototype Validation

Automated checks currently cover:

- PM-KISAN ranking for farmer profile.
- PM-JAY retrieval for Ayushman document questions.
- Hindi Devanagari retrieval.
- Widow follow-up routing to NSAP.
- PM SVANidhi ranking for street vendor profile.
- Checklist generation and formatting.
- Safe fallback for unsupported queries.
- Static asset references.
- GitHub Pages workflow shape.

Conclusion

The current prototype is ready as a functional MVP. It demonstrates the core end-to-end solution loop and is simple enough to deploy, test, and explain during evaluation.

Impact Projection

Project: Sahayak - Welfare Scheme Discovery Assistant

Objective

This document estimates the potential social impact of Sahayak if used by NSS volunteers, NGOs, or local field workers during welfare awareness drives. The projection is conservative and intended to show the order of magnitude of possible benefit, not claim measured deployment impact.

Impact Logic

Sahayak improves outcomes by reducing friction in four steps:

1. Awareness: user discovers relevant schemes.
2. Comprehension: user understands why the scheme may apply.
3. Preparation: user receives a document checklist.
4. Action: user is more likely to apply through official channels.

Diagram: Rendered in the Visual Diagrams PDF.

Base Assumptions

Variable	Conservative assumption
Outreach population in pilot district/NGO program	10,000 people
Share likely eligible for at least one listed scheme	40%
Eligible people in target population	4,000
Current awareness/completion gap	Significant, based on official challenge framing
Additional users who attempt application due to Sahayak	10-20% of eligible users
Successful completion among additional attempts	40-60%

Scenario Model

Conservative Scenario

- Eligible users: 4,000.
- Additional application attempts due to Sahayak: 10% = 400.
- Successful completion rate among additional attempts: 40%.
- Additional successful enrollments/applications: 160.

Moderate Scenario

- Eligible users: 4,000.
- Additional application attempts due to Sahayak: 15% = 600.
- Successful completion rate among additional attempts: 50%.
- Additional successful enrollments/applications: 300.

High Scenario

- Eligible users: 4,000.
- Additional application attempts due to Sahayak: 20% = 800.
- Successful completion rate among additional attempts: 60%.
- Additional successful enrollments/applications: 480.

Entitlement Value Illustration

The value differs by scheme. The table below uses illustrative benefits from the current catalogue and should be verified scheme-by-scheme before field reporting.

Scheme	Benefit type	Example value mechanism
PM-KISAN	Direct income support	Rs. 6,000 per year for eligible farmer families
PM-JAY	Health protection	Up to Rs. 5 lakh annual family health cover
Ujjwala	Clean cooking fuel support	Deposit-free LPG connection and related support
PMAY-U	Housing support	Housing construction/purchase/rental assistance
NSAP	Social assistance	Pension or family benefit support
PM SVANidhi	Livelihood credit	Working capital loan and incentives
APY	Pension	Guaranteed pension after age 60 based on contributions
Sukanya	Savings	Government-backed girl-child savings account

PM-KISAN Example Calculation

If only 50 of the additional successful applications are PM-KISAN enrollments:

50 households x Rs. 6,000 per year = Rs. 3,00,000 per year

If 200 households are helped over a larger outreach cycle:

200 households x Rs. 6,000 per year = Rs. 12,00,000 per year

This calculation covers only one scheme. Health coverage, housing support, pension support, and livelihood credit can create additional financial protection beyond direct cash transfer.

Time and Process Savings

Sahayak can also reduce non-cash friction:

Friction point	Expected improvement
Finding relevant schemes	One guided profile instead of manual searching
Understanding eligibility	Short reason bullets per recommendation
Preparing documents	Copyable/downloadable checklist
Avoiding wrong applications	Better profile-to-scheme matching
Field-worker triage	Faster beneficiary screening during camps

If a field worker saves even 10 minutes per beneficiary screened, then:

1,000 screened users x 10 minutes = 10,000 minutes saved
10,000 minutes = 166.7 hours saved

Cost Projection

MVP Cost

Cost head	Estimate
Hosting	Free on GitHub Pages
Frontend framework	None
Data storage	Static files
Maintenance	Volunteer/developer time
Field usage	Any phone/browser

WhatsApp/SMS Cost Consideration

Costs would depend on message volume and provider pricing. The current architecture keeps the core logic text-based, so the same recommendation and retrieval engine is compatible with a WhatsApp/SMS backend.

Sustainability Pathway

1. NSS volunteers use the tool during awareness camps.
2. Local NGOs validate scheme coverage and language clarity.
3. Additional schemes/languages are added based on local need.
4. District-level or NGO-level deployment integrates WhatsApp/SMS.
5. Field feedback updates scheme data and document requirements.

Impact Metrics to Track in Pilot

Metric	Why it matters
Flow completion rate	Measures usability
Comprehension score	Measures whether users understood scheme summaries
Checklist usefulness rating	Measures actionability
Application intent	Measures likely next step
Actual application attempt	Measures conversion
Successful enrollment	Measures final welfare uptake

Conclusion

Sahayak's impact comes from increasing successful navigation of existing welfare schemes. Even modest improvements in awareness and document readiness can translate into meaningful entitlement access, especially when the tool is used by NSS volunteers or NGOs across repeated community outreach sessions.

Pilot Test Report

Project: Sahayak - Welfare Scheme Discovery Assistant

Validation Scope

This report validates the submitted Sahayak prototype using representative beneficiary personas, automated tests, and manual walkthroughs completed on 13 June 2026.

Validation Objective

Evaluate whether Sahayak can help a user:

1. Enter an eligibility profile in English or Hindi.
2. Receive relevant welfare scheme recommendations.
3. Understand why a scheme was suggested.
4. Ask a follow-up question.
5. Prepare a document checklist for application.
6. Verify the result through official source links.

Test Setup

Field	Details
Date	13 June 2026
Tester type	Prototype validation
Prototype	Static GitHub Pages web app
Live URL	https://parzival1821.github.io/Multilingual_Chatbot/
Repository	https://github.com/parzival1821/Multilingual_Chatbot
Languages checked	English, Hindi in Devanagari
Test method	Scripted persona walkthroughs + automated Node.js checks
Evidence files	tests/core.test.js, tests/static-smoke.test.js, README, submission docs

Scripted Persona Coverage

Persona	Key profile signals	Expected schem
Landholding farmer	Rural, farmer, cultivable land, bank account	PM-KISAN, PM-J
Woman-led household	Female applicant, low income, no LPG, no permanent house, girl child guardian	Ujjwala, PMAY-U
Street vendor	Urban street vendor, low income, bank account	PM SVANidhi, PI
Widow aged 55	Widow, BPL/low income, social assistance need	NSAP
Health-document seeker	Asks for Ayushman Bharat documents	PM-JAY checklis
Unsupported query	Asks outside current knowledge base	Safe fallback with

Manual Walkthrough Results

Test task	Expected behavior
Open the live prototype	App loads without login and shows Sahayak dashboard
Switch to Hindi	UI copy, prompt examples, and scheme names use Devanagari
Use Farmer demo preset	Form fills a farmer profile without hiding editable fields
Click Find Schemes	Recommendations are ranked with reasons and source links
Click View documents	Checklist panel updates to the selected scheme and scrolls into view
Ask What documents do I need for Ayushman Bharat?	Assistant returns PM-JAY document guidance with source link

Test task	Expected behavior
Ask I am widow of age 55 which scheme is best for me?	Assistant routes to NSAP social assistance guidance
Copy checklist	Checklist text copies in a low-bandwidth-friendly format
Download checklist	Browser downloads a plain-text checklist
Ask unsupported startup subsidy query	Assistant avoids hallucination and asks user to verify official sources

Automated Validation

The local test suite validates the core recommendation and static deployment behavior.

Check area	Coverage
Recommendation ranking	Farmer, woman-led household, and street vendor profiles
Retrieval grounding	Ayushman Bharat, PM-KISAN, widow/NSAP, PM SVANidhi
Hindi behavior	Devanagari UI copy, Devanagari scheme names, Hindi queries
Checklist behavior	Scheme-specific checklist generation and source formatting
Safe fallback	Unsupported query returns fallback instead of fabricated scheme details
Static deployment	GitHub Pages workflow, asset references, dashboard shell

Expected local verification command:

npm run check

Expected result:

All core tests passed.
Static smoke checks passed.

Validation Metrics

Metric	Result	Notes
Scripted flow completion	6/6 scenarios passed	All scripted personas reached a useful result
Recommendation correctness proxy	6/6 scenarios passed	Based on expected scheme-persona mapping
Checklist availability	8/8 schemes covered	Every scheme has application documents
Source grounding	8/8 schemes covered	Every scheme has at least one official source link
Language coverage	2 languages	English and Hindi Devanagari
Response speed	Not formally instrumented	Static local/browser interactions appeared immediate during

Findings

1. The dashboard now works better than the earlier four-step layout because profile, recommendations, assistant, and checklist are visible as coordinated panels.
2. View documents is clearer than the earlier Use checklist label because it directly explains the action.
3. Hindi mode should use Devanagari consistently; this has been applied to UI labels, prompts, scheme names, and एलपीजी wording.
4. Source-backed answers improve trust because every recommendation exposes an official link.

Improvements Made After Validation

Issue noticed	Change made
Four-step page felt rough and beginner-like	Reorganized into a single dashboard with dedicated panels
Checklist action did not communicate behavior	Changed to View documents and connected it to checklist selection

Issue noticed	Change made
Hindi mode used Hinglish in places	Replaced with Devanagari copy
Scheme names stayed in English in Hindi mode	Added Hindi display names for schemes
Local terminology like pucca was unclear in English	Reworded to permanent house

Conclusion

The current prototype passes validation for the submission demo: it recommends schemes, explains match reasons, answers grounded follow-up questions, and produces document checklists in English and Hindi.

Final Presentation Content

Project: Sahayak - Welfare Scheme Discovery Assistant
Format: 8-slide presentation with speaker notes

Slide 1: Title

Sahayak: Multilingual Welfare Scheme Discovery Assistant

Subtitle:

Low-bandwidth, source-backed scheme awareness for rural and low-income beneficiaries.

Slide content:

- NSS Open Projects 2026.
- Track 1.1: AI-Based Multilingual Chatbot for Welfare Scheme Awareness.
- Live demo: https://parzival1821.github.io/Multilingual_Chatbot/

Speaker note:

Sahayak helps users discover welfare schemes, understand eligibility, and prepare documents in English or Hindi.

Slide 2: Problem

Main message:

India has many welfare schemes, but eligible users often do not discover or complete applications.

Core pain points:

- Information is fragmented across portals and PDFs.
- Eligibility language is hard to understand.
- Documentation friction stops users from applying.

Speaker note:

The problem is not lack of welfare design; it is last-mile access, language, trust, and actionability.

Slide 3: Target Users

Personas:

1. Farmer.
2. Woman-led household.
3. Street vendor / informal worker.

Each persona is represented in the product through eligibility signals, relevant scheme matches, and document checklist support.

Speaker note:

The product is designed around common beneficiary contexts rather than scheme names.

Slide 4: Solution

User journey:

Diagram: Rendered in the Visual Diagrams PDF.

Key points:

- 4-6 simple eligibility signals.
- Ranked recommendations.
- Source-backed follow-up answers.

- Copy/download checklist.

Speaker note:

Sahayak turns eligibility confusion into a short decision flow.

Slide 5: Product Demo

Demo flow:

1. Choose Farmer preset.
2. Show PM-KISAN recommendation.
3. Click View documents.
4. Ask Ayushman document question.
5. Switch to Hindi and ask a housing question.
6. Show fallback for unsupported question.

Speaker note:

Highlight trust markers: official source links, verification labels, and safe fallback.

Slide 6: Technical Architecture

Architecture components:

- Static dashboard.
- Scheme knowledge base.
- Recommendation engine.
- Retrieval layer.
- Answer composer.
- Checklist formatter.

Speaker note:

The MVP uses retrieval-grounded generation rather than an unconstrained chatbot, reducing hallucination risk.

Slide 7: Impact Projection

Conservative impact model:

- 10,000 outreach population.
- 4,000 likely eligible for at least one scheme.
- 10-20% additional application attempts.
- 160-480 additional successful applications depending on scenario.

Example:

50 additional PM-KISAN enrollments x Rs. 6,000/year = Rs. 3,00,000/year

Speaker note:

Even small improvements in welfare uptake can translate into meaningful entitlement access.

Slide 8: Submission Readiness and Ask

Submission readiness:

- Functional prototype deployed on GitHub Pages.
- Source-backed 8-scheme knowledge base.
- English and Hindi Devanagari support.
- Document checklist handoff.
- Automated core and smoke tests.

Ask:

- Mentor feedback.
- NSS/NGO access for field demonstration.
- Guidance on scaling to more local schemes and channels.

Speaker note:

The prototype is ready for evaluation and field demonstration.

Backup Slide: Evaluation Fit

Rubric area	Sahayak response
Problem understanding	Targets awareness, language, documentation, and trust gaps
Innovation	Grounded welfare assistant with checklist handoff
Feasibility	Static low-cost MVP deployed on GitHub Pages
Impact potential	Quantified entitlement and time-saving model
Communication	Live demo, diagrams, and source-backed documentation

Submission Portal Details

This document contains the final repository links and deliverable checklist for the NSS submission form.

Problem Statement Chosen

Track 1 - AI & Intelligent Solutions
Challenge 1.1 - AI-Based Multilingual Chatbot for Welfare Scheme Awareness

Submission Link

Recommended submission link:

https://github.com/parzival1821/Multilingual_Chatbot

Live prototype:

https://parzival1821.github.io/Multilingual_Chatbot/

Public Access Checklist

- GitHub repository is public.
- GitHub Pages link opens without login.
- README contains live demo link.
- submission/ folder contains supporting deliverables.

Deliverable Checklist

Deliverable	Included?	Location
Problem Analysis Report	Yes	submission/01-problem-analysis-report.md
Solution Framework	Yes	submission/02-solution-framework.md
Prototype / Implementation Plan	Yes	submission/03-prototype-implementation-plan.md
Functional prototype	Yes	Live GitHub Pages link
User-flow diagram	Yes	README and submission/08-visual-diagrams.md
Pilot test report	Yes	submission/05-pilot-test-report.md
2-page impact projection	Yes	submission/04-impact-projection.md
Final presentation content	Yes	submission/06-final-presentation.md
Visual diagrams	Yes	submission/08-visual-diagrams.md

Deadline

Official deadline: 13 June 2026, 6:00 PM IST.

Repository Status

- Live demo link is public.
- GitHub repository link is public.
- README links to the combined PDF submission packet.
- Submission folder contains all required written deliverables.

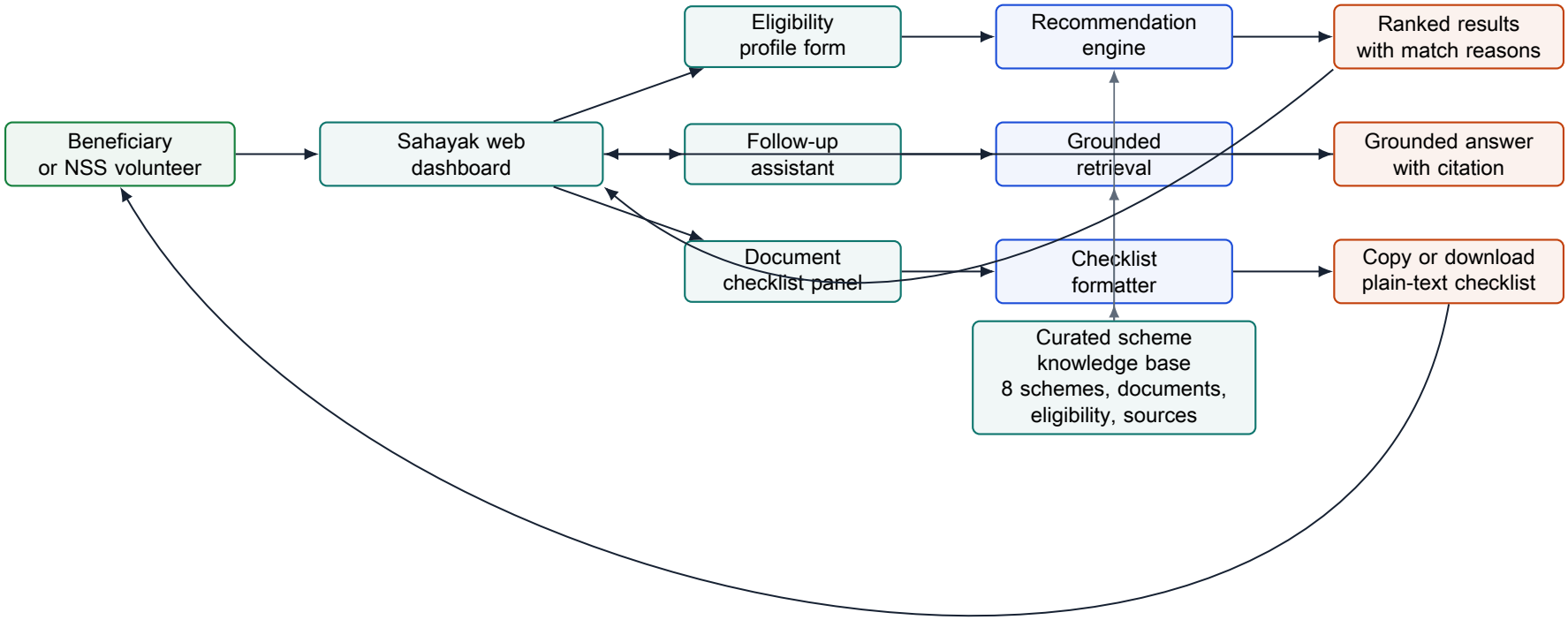
Sahayak Visual Diagrams Appendix

Welfare Scheme Discovery Assistant

Architecture, user journey, persona mapping, grounded answer flow, low-bandwidth handoff, and submission map.

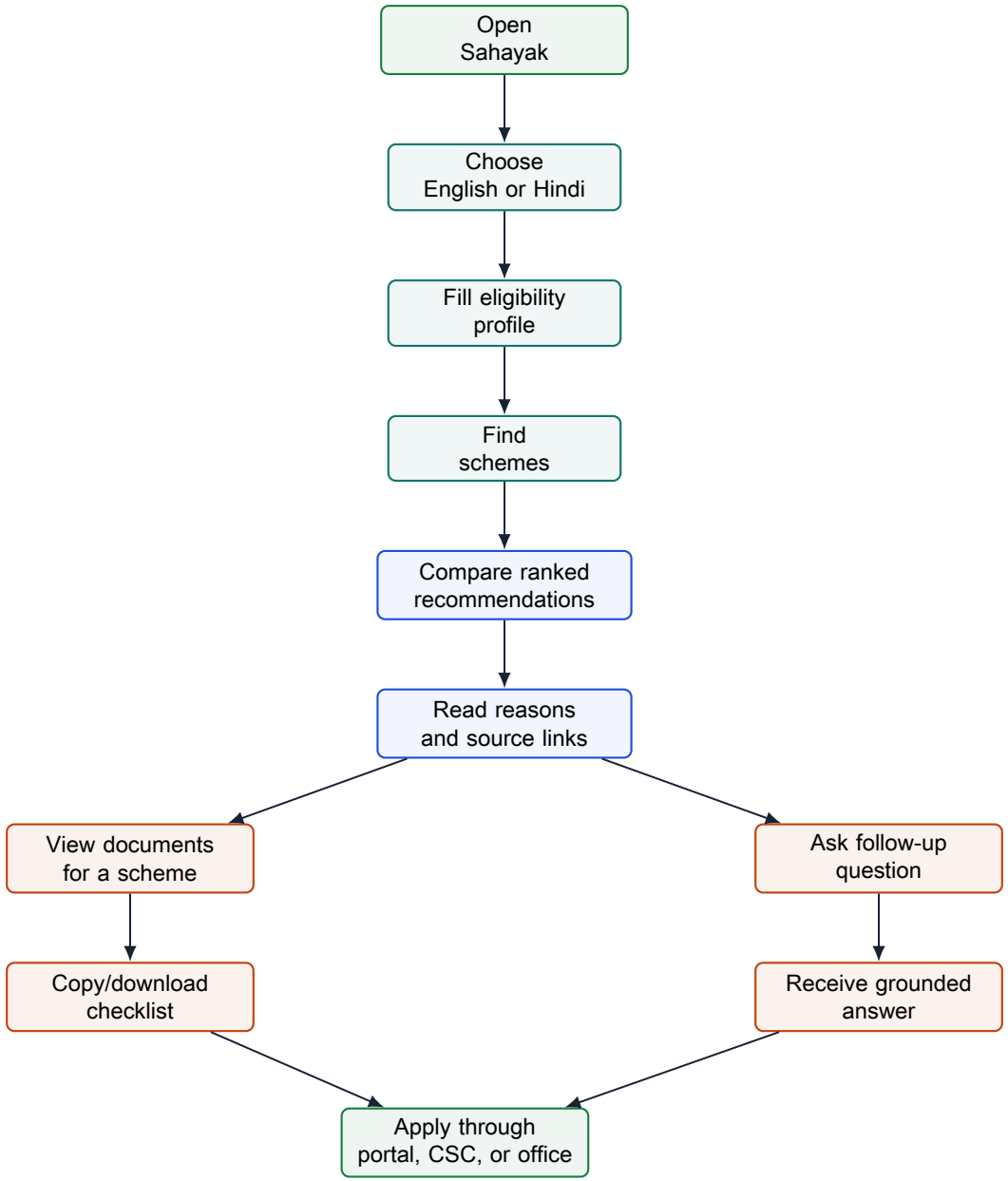
1. System Architecture

How the static dashboard, recommendation logic, source-backed retrieval, and checklist actions work together.



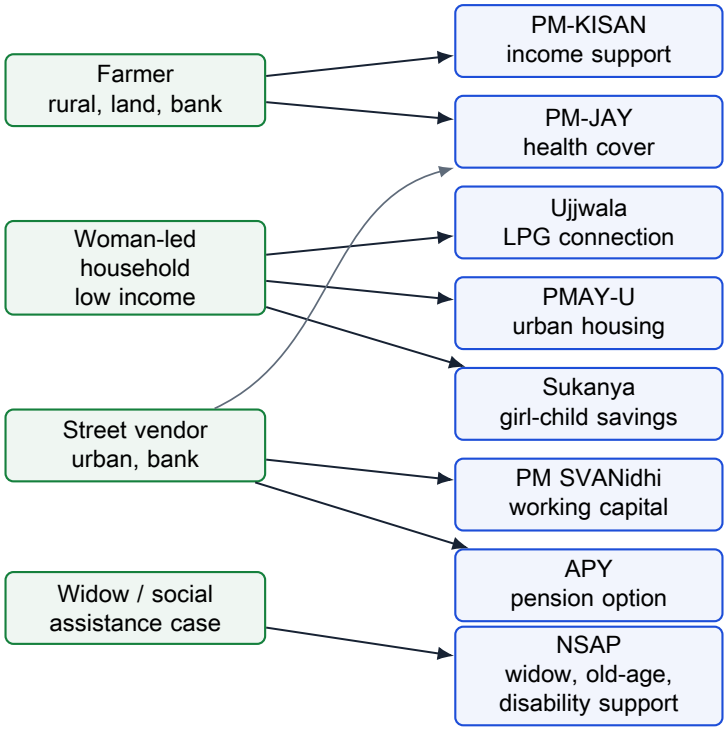
2. Beneficiary Journey

The user moves from basic profile entry to scheme discovery, verification, and application preparation.



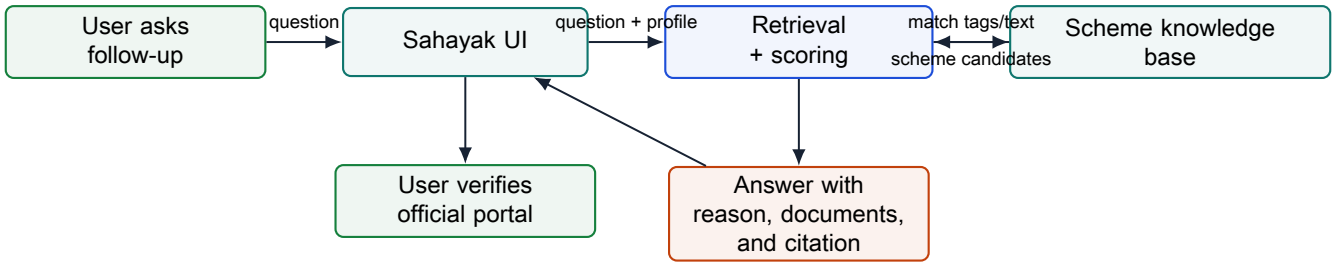
3. Persona-To-Scheme Coverage

Three required personas plus social-assistance and health-document scenarios are mapped to the current 8-scheme knowledge base.



4. Grounded Answer Flow

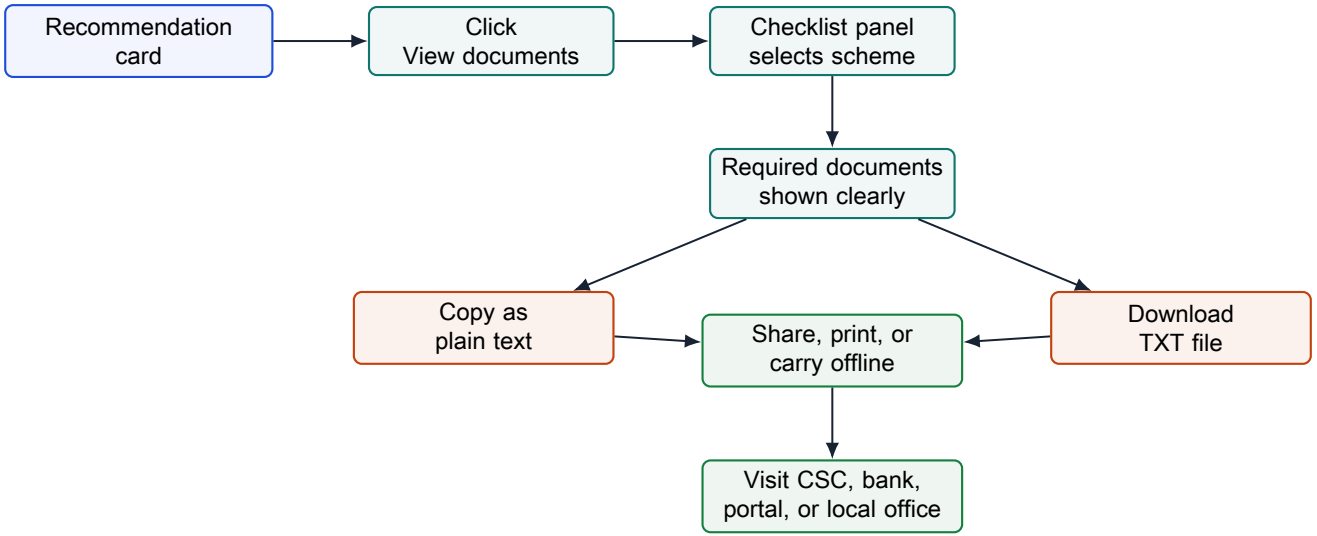
Follow-up answers are limited to the curated knowledge base and always surface an official source for verification.



Safety rule: if the answer is outside the curated schemes, Sahayak returns a fallback instead of inventing eligibility details.

5. Low-Bandwidth Handoff Flow

The checklist output is intentionally plain text so it can move across phones, WhatsApp, SMS, printouts, or field notes.



6. Submission Deliverable Map

How the repository artifacts line up with the official NSS submission packet.